Disaster Network

Crisis operations, at scale

An operations benchmark for a flood-response portfolio near Marseille.

01 THE CRISIS WORKLOAD

On 20 October 2026, flooding around Istres and Berre Lagoon brought evacuations, reception centres and closed access roads. Météo-France reported episode totals of 275 mm in Istres and 315 mm in Saint-Chamas [1, 2].

Evaluation portfolio: 80 households / 170 people. The operating model below assumes one initial call, one claim form and three supporting files per household. These volumes describe the selected portfolio, not an official event-wide claims count.

80
initial hotline calls

80
claim forms

240
supporting files

Workstream	Effort / claim	Scenario volume	With Disaster Network
Initial call	12-18 min	80 calls	Voice intake and structured report
Form + policy lookup	5-7 min	80 forms	Verified dossier and clause retrieval
Three-file review	9-14 min	240 files	Photo/PDF extraction and checks
Coordination + status	4-6 min	80 dossiers	Priorities, teams and one shared view

02 MODELLED OPERATING ECONOMICS

40-60 h
manual staff effort

8-16 h
assisted staff effort

~25-50 h
staff capacity released

Initial handling + planning	Manual	With Disaster Network
Human effort / claim	30-45 min	6-12 min
Operating cost / claim	~EUR 20-45	~EUR 5-15
Operating budget / 80 claims	~EUR 1.5-4k	~EUR 0.4-1.2k
At 1,000 claims, same assumptions	~EUR 20-45k	~EUR 5-15k

About 25-50 staff hours released across an 80-claim surge.

Planning assumptions: EUR 40-60 per staff hour; 2-4 minutes of review for every claim, plus 20-40% escalated for 20 extra minutes; EUR 1-3 technical allowance per claim. Ranges are scenario estimates to validate in a pilot. Scope: initial handling and assistance planning; indemnities, field services, integration, subscription and maintenance are budgeted separately.

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Controlled crisis operations

03 HOW THE HARNESS CONTROLS RISK

Operational risk	Implemented control	Result for the workflow
Wrong dossier or premature access	Policy number, first name and stored challenge before private reads	Identity gate before contract access
Unsupported policy or invoice conclusions	Linked clauses/PDF pages; structured extraction; server checks on dates, amounts and limits	Evidence-linked proposals and explicit missing information
Unavailable team or unsafe access	Skill, service area, availability, free capacity and access checks	Ineligible assignments held back
Duplicate or outdated action	Idempotent report save, source hashes and proposal freshness check	Repeat requests and changed dossiers are controlled
Unreviewed insurer decision	Explicit handler review before recording eligible planned assignments	Human decision retained with the proposal and tool history

04 THE INSURER SEES THE WHOLE RESPONSE

One map and queue: households, urgency, habitability, access and outstanding needs. **One dossier:** policy, evidence, missing information and decision rationale. **One resource view:** provider skills, remaining capacity, assignments and replanning. Financial views keep estimates, commitments and paid amounts distinct.

Connectors: OpenAI voice, document reading and classification; Aiven PostgreSQL; Géorisques GASPAR; optional official-domain web search. External bookings and payments remain separate integrations.

05 MEASURED RESULT: CAPACITY-AWARE REPLANNING

Delta Drying becomes unavailable. Both approaches receive the same two affected missions and provider directory. The baseline picks the first available matching provider; the application also checks remaining capacity, access and reservations within the plan.

Recorded result	Simple allocation	Application plan
Feasible replacements	0 / 2	2 / 2
Capacity conflicts	2	0
Existing blocked missions preserved	-	3 / 3

Both disrupted missions are reassigned; unrelated missions remain unchanged. This recorded execution measures the planning rules. The cost model on page 1 is calculated separately; model classification accuracy is a separate evaluation.

METHOD & SOURCES

Planning run: 22 October 2026, 13:14 Europe/Paris. Reproduce with `bun run scripts/submission-benchmark.js`. Raw results and source/dataset hashes: benchmark-results.json/.csv. Workload formulas and assumptions: operations-model.json. Guardrails above were reviewed in the current source.

[1] Météo-France: Mediterranean rainfall episode, 21 October 2026. [2] City of Istres: emergency plan and evacuation updates, 20 October 2026.